

TLB misses and lesser RAM usage for page tables. However, hugepages carry the same restriction as with physical servers—if enabled, guest pages can't be swapped or ballooned.

- Enables hosts to discover new SAN storage, issue NPIV operations and present SAN storage to guests.
- Interfaces with netcf. netcf can configure bridges and network devices and pass them on to NetworkManager and libvirt. This will ease future integration of all network configuration in one place and allow one to use NetworkManager and place guests on bridges without hand-editing configuration files.
- Use cgroups to set resource limits on VMs
- Pin vcpus to physical CPUs: since VMs running under KVM are normal Linux processes, this can also be done via the command line.
- Support for creation of encrypted qcow2 images
- Compressed save image format
- Better device assignment support
- Clone storage used for guests

There are other features available as well, which don't involve KVM. Some of them include VirtualBox support, VMware ESX hypervisor support, Power Hypervisor support, etc.

The virt-manager UI has been greatly improved and leads to a better overall user experience. A few key features that are now available are:

- Improved UI, overhaul of the main manager GUI
- CPU pinning support
- System tray icon for easier access to VMs
- Ability to view and change security settings
- New VM clone wizard

Other improvements

In addition to the features mentioned above, there's also a switch from the etherboot option ROM to the gPXE-based option ROM. This allows a more stable network boot experience for virtual machines. This

Ready reference table

Security	▪ QEMU processes started with lower privileges
Performance	▪ MSIX ▪ MSI-X support for assigned devices ▪ SR-IOV ▪ Huge-page backed memory ▪ KSM ▪ Qcow2 performance enhancements, stabilization
Developers	▪ qdev device model ▪ qemu-io ▪ Guest debugging
Stability	▪ Stable PCI bus addresses (Guest ABI)
Usability	▪ Pointing to iso images over http for guest installation ▪ Allow setting maximum downtime for migration ▪ NIC hotplug ▪ Automatic guest bridging with the netcf library ▪ Network booting with gPXE ▪ libguestfs for offline guest image viewing / editing

makes sense, because most etherboot development has moved to the gPXE project. Fedora will receive a more modern boot infrastructure, as well as quicker bug fixes.

Newly-introduced tools

All talk so far has revolved around new features that are available while actually running the virtual machines; now, a new set of tools and utilities has been added for examining guests when they're offline.

libguestfs is a library that allows access to and modification of guest disk images. guestfish is an interactive shell tool for editing guest disk images.

These tools allow users to perform a host of activities on guest disk images:

- Make batch configuration changes
- View and edit files inside guests without mounting the disk image
- Getting used/free stats for guest disks
- Performing partial guest backups
- Performing partial guest clones
- Changing settings/registry entries, etc
- Migrating between various virtualisation systems (physical to

virtual, Xen to KVM, etc)
libguestfs can connect to libvirt, so existing guests can be directly referenced to within the guestfish shell. It also understands a host of file system formats—for example, ext2/3/4, btrfs, XFS, FAT, NTFS, LVM, etc.

The features presented here are not exhaustive. Moreover, there has been a lot of bug fixing in each of the projects mentioned, which makes the entire release very compelling for anyone to try out the new features and see where the future of virtualisation is headed. 

References

- Upstream project pages:
- Linux: <http://kernel.org>
- KVM: <http://www.linux-kvm.org>
- QEMU: <http://www.qemu.org>
- libvirt: <http://libvirt.org>
- virt-manager: <http://virt-manager.org>
- libguestfs: <http://libguestfs.org>
- Fedora 12 features and history of Fedora Virtualization: <https://fedoraproject.org/wiki/Virtualization/History>
- Interviews with some developers about new virtualisation features in Fedora 12: http://www.fedoraproject.org/wiki/Virtualization_improvements_in_Fedora_12

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